

QXU6033 Advanced Polymer Chemistry

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1. Introduction to Module

■ Assessment

Activity	Score
Professional Practice Evaluation(supervision)	20%
Coursework	30%
Final Exam	50%

1.1. Study Goal

■ Controlled Radical Polymerisation

Mechanistic principles of controlled / living radical polymerisation (CRP)
 Kinetic control, equilibrium processes, and radical deactivation strategies
 Molecular weight control, dispersity ($\bar{M}_w/\bar{M}_n$), and end-group fidelity

■ Core CRP methodologies

Nitroxide-Mediated Polymerisation (NMP)
 Reversible Addition–Fragmentation Chain Transfer (RAFT)
 Atom Transfer Radical Polymerisation (ATRP)

■ Advanced macromolecular architectures

Self-Condensing Vinyl Polymerisation (SCVP)
 Strathclyde synthesis and hyperbranched structures
 Star polymers and multi-arm architectures

■ From synthesis to structure

Block copolymer design and synthesis
 Microphase separation and self-assembly

Introductory Discussions I

Q: A has a structure that enables it to polymerise through chain-growth polymerisation, while B is step-growth.

Scenario 1: Imagine you start polymerising A, and half-way through the polymerisation, you add C, also another monomer, which can polymerise through chain-growth polymerisation, and, that can polymerise together well with A (this is called “copolymerise”). Separately, imagine Scenario 2: Use B instead A, then add D, a monomer, which can polymerise through step-growth polymerisation, and, that can copolymerise well with B. At the end of both polymerisation processes, what type of polymer samples will you have in each Scenario?

A: S1: AAAAAA-(random A and C)-(if one is finish, pure another one).

S2:(block B)-some D-(block B) and cycle B.

2. Controlled Radical Polymerisation: Introduction

Note

CRP, controlled / living radical polymerisation.

Also Reversible Deactivation Radical Polymerisation, RDRP.

2.1. Towards Living Polymerisation

As long as “living” polymer is kept tightly closed (no H₂O, ROH, CO₂, O₂) more monomer can be added and the molecular weight will increase.

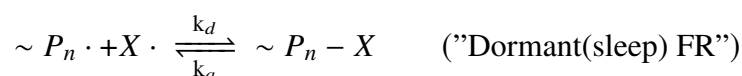
1. MW increases steadily with time/conversion
2. Narrow PDI (< 1.2 or 1.3)
3. Can add more monomer or another monomer/blocks.

■ Kinetic Issue

To make FRP “living”, need to inhibit termination and chain-transfer without affecting initiation/propagation too much.

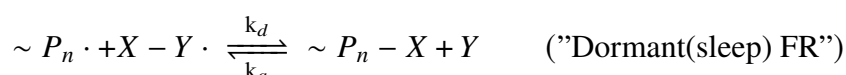
■ Strategy 1: Reversible Termination

Also, Reversible Deactivation.



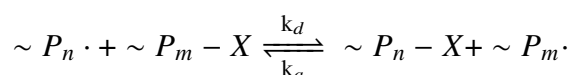
Note: Do not use Strategy 1, either refer to specific chemistry (NMP, or TEMPO) or “reversible deactivation”

■ Strategy 2: Catalytic/Assisted Reversible Termination



X can be single atom transferred to P·, Y normally a transition metal.

■ Reversible Chain-Transfer(Degenerative Transfer)



In processes $\sim P_n \cdot$ or $\sim P_m \cdot$ themselves, **Termination still occurs.**

To be effective

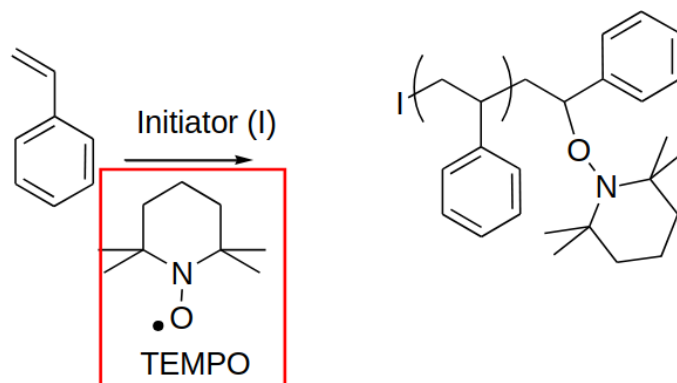
Reversible Termination should be $k_d > k_a$

Reversible Chain-Transfer is $k_d \sim k_a$

■ Controlled Radical Polymerisation by Reversible Deactivation

Specific chemistries:

Nitroxyl radicals (e.g. TEMPO) for Nitroxyl Mediated Polymerisation (NMP)



TEMPO: (2,2,6,6-Tetramethylpiperidin-1-yl)oxyl radical

Note

The equilibrium with TEMPO doesn't happen necessarily at every single monomer addition.